MSBA 402 Practices Discussion 1

University Enrollment System

Below are three tables from a database schema used by a university:

Students Table

student_id	student_uuid	email	full_name	major	advisor_id
1001	8f9a2b10-1111	a.smith@univ.edu	Alice Smith	Computer Science	501
1002	8f9a2b10-2222	b.jones@univ.edu	Bob Jones	Data Analytics	502
1003	8f9a2b10-3333	c.davis@univ.edu	Charlie Davis	Computer Science	501
1004	8f9a2b10-4444	d.miller@univ.edu	Diana Miller	Business Analytics	503

Advisors Table

advisor_id	advisor_name	office_building	office_room	department
501	Dr. Aris Thorne	Science Hall	302	Computer Science
502	Dr. Elena Rostova	Business Hall	104	Analytics
503	Dr. Marcus Vance	Business Hall	108	Business

Enrollments Table

enrollment_id	student_id	course_code	semester	registration_time	final_grade
50001	1001	CS-101	Fall 2026	2026-08-15 09:00:00	A
50002	1001	MSBA-302	Fall 2026	2026-08-15 09:05:00	A-
50003	1002	MSBA-302	Fall 2026	2026-08-15 10:12:00	B+
50004	1003	CS-101	Fall 2026	2026-08-16 11:30:00	B

Questions

Question 1

Which four fundamental operations define basic data interaction within a relational database system (often summarized by the acronym CRUD)?

- A) Create, Remove, Update, Destroy
- B) Create, Read, Update, Delete
- C) Compile, Read, Upload, Delete
- D) Create, Retrieve, Unlink, Discard

Question 2

Based on the **Students** table in the scenario above, which column best functions as a **Surrogate Key**?

- A) email
- B) full_name
- C) student_id
- D) major

Question 3

What is the primary role of a **Foreign Key** in a database design?

- A) To automatically sort rows in ascending order based on numeric values.
- B) To allow two tables to be physically merged into one single table.
- C) To enforce referential integrity rules by linking records in a child table to valid records in a parent table.
- D) To uniquely identify every single user accessing the database application.

Question 4

In the **Enrollments** table, which column acts as a **Foreign Key** referencing the parent table Students?

- A) enrollment_id
- B) student_id
- C) course_code
- D) registration_time

Question 5

Consider the **Students** table. If the database administrator deletes the record for advisor_id = 501 from the **Advisors** table without updating or removing related student records, what condition occurs in the Students table?

- A) The database automatically converts all foreign keys into primary keys.
- B) Related records in the Students table become orphaned (lacking a valid parent reference).
- C) The Students table will be automatically dropped from the database schema.
- D) A primary key constraint violation will trigger on the Students.student_id column.

Question 6

Which of the following defines a **Candidate Key**?

- A) A primary key from another table used to establish referential integrity.
- B) A column or minimal set of columns that can uniquely identify every single row in a table.
- C) An artificially generated key created by the database system that has no real-world business meaning.
- D) A human-readable key derived from real-world business attributes used intuitively by end users.

Question 7

In the **Students** table, assuming no two students share the same university email, which of the following is a valid candidate key set?

- A) {student_id}, {email}, {student_uuid}
- B) {major}, {advisor_id}
- C) {full_name}, {major}
- D) {advisor_id}, {email}

Question 8

Which key type is meaningful in the real-world business context and easily recognizable to external users (e.g., a natural identifier like a Social Security Number or a domain email address)?

- A) Surrogate Key
- B) Natural Key
- C) Foreign Key
- D) Composite Schema

Question 9

In database modeling, what term refers to the overall structure, organization, and design of the database (or table structures, data types, and constraints)?

- A) Tuple
- B) Schema
- C) Record
- D) Transaction

Question 10

Look at the following schema specification rule:

The registration_time column in Enrollments uses data type DATETIME and is NOT NULL.

What does the NOT NULL constraint ensure?

- A) The column must only contain unique numbers.
- B) The column cannot be left empty or missing when inserting a record.
- C) The column value must match an entry in another table.
- D) The column value cannot be updated once created.

Solutions & Answer Key

Question 1

- **Correct Answer: B) Create, Read, Update, Delete**
- **Explanation:** CRUD represents the four basic functions of persistent storage: **C**reate, **R**ead, **U**pdate, and **D**elete.

Question 2

- **Correct Answer: C) student_id**
- **Explanation:** A surrogate key is an artificially generated identifier created by the database (often an integer or auto-incremented ID) that carries no inherent real-world business meaning.

Question 3

- **Correct Answer: C) To enforce referential integrity rules by linking records in a child table to valid records in a parent table.**
- **Explanation:** Foreign keys are used to enforce referential integrity constraints, ensuring that relationships between tables remain valid and preventing invalid data references. (Note: Foreign keys do *not* merge tables).

Question 4

- **Correct Answer: B) student_id**
- **Explanation:** student_id in the Enrollments table references the student_id primary key in the parent Students table.

Question 5

- **Correct Answer: B) Related records in the Students table become orphaned (lacking a valid parent reference).**
- **Explanation:** Removing a parent record (Advisors) without handling referencing child records (Students) leaves the child records without a matching parent key, making them "orphaned".

Question 6

- **Correct Answer: B) A column or minimal set of columns that can uniquely identify every single row in a table.**
- **Explanation:** A candidate key is any column or minimal set of columns capable of uniquely identifying each row in a table; one candidate key is selected as the Primary Key.

Question 7

- **Correct Answer: A) {student_id}, {email}, {student_uuid}**
- **Explanation:** Each of these individual attributes uniquely identifies a student in the table without duplicates, making each an independent candidate key.

Question 8

- **Correct Answer: B) Natural Key**
- **Explanation:** A Natural Key is formed from real-world business attributes that have logical meaning to human users (e.g., email address, order reference number).

Question 9

- **Correct Answer: B) Schema**
- **Explanation:** Schema refers to the overall design and logical organization of the database or table structures.

Question 10

- **Correct Answer: B) The column cannot be left empty or missing when inserting a record.**
- **Explanation:** A NOT NULL constraint enforces that a value must be supplied for that column, preventing missing or null entries.